Algorithm Analysis Fundamentals

Time complexity analysis of iterative and recursive algorithms; verify Big-O, Ω, Θ for sample programs

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DAA Topic 01 Experiments

Experiment 1: Linear Search (Iterative)

● Objective: To analyse time complexity of linear search.

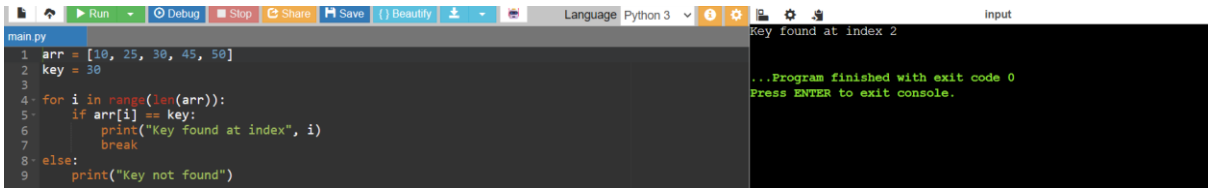
● Problem Statement: Write a Python program to search for a key in an array using linear search.

● Sample Input: [10, 25, 30, 45, 50], Key = 30

● Sample Output: Key found at index 2

● Complexity: Ω: O(1), O: O(n), Θ: Θ(n)

● Explanation: Demonstrates sequential scanning and inefficiency for large inputs.



Experiment 2: Binary Search (Recursive)

● Objective: To compare recursive binary search complexity.

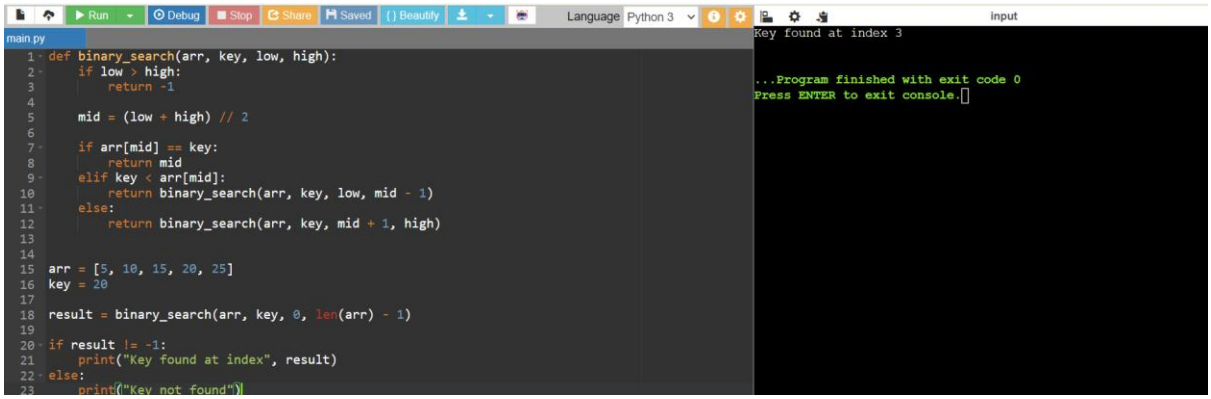
● Problem Statement: Implement recursive binary search in Python.

● Sample Input: [5, 10, 15, 20, 25], Key = 20

● Sample Output: Key found at index 3

● Complexity: Ω: O(1), O: O(log$_{f0}$n), Θ: Θ(log$_{f0}$n)

● Explanation: Shows divide-and-conquer and logarithmic efficiency.



Experiment 3: Factorial (Iterative vs Recursive)

● Objective: To compare iterative and recursive factorial computation.

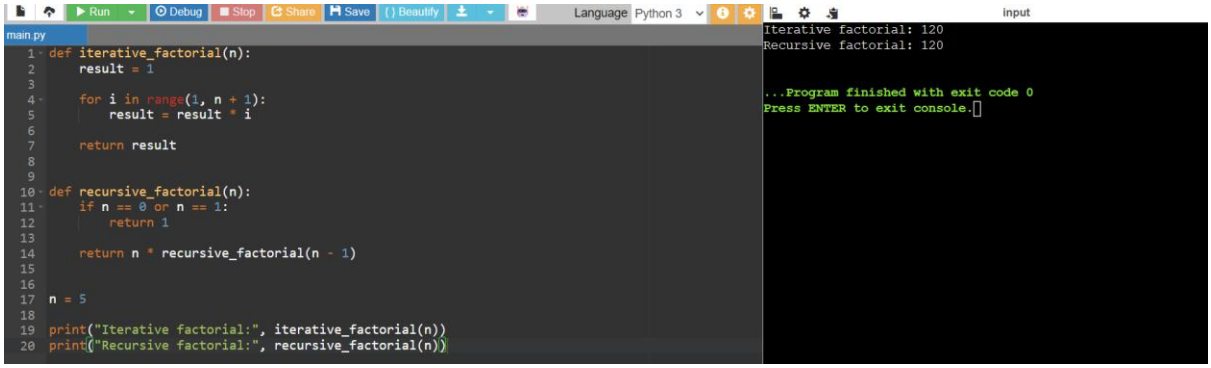
● Problem Statement: Write programs to compute factorial using both methods.

● Sample Input: n = 5

● Sample Output: 120

● Complexity: Both O(n)

● Explanation: Highlights recursion depth vs iterative loops.



Experiment 4: Fibonacci Series

● Objective: To analyze recursive vs iterative Fibonacci generation.

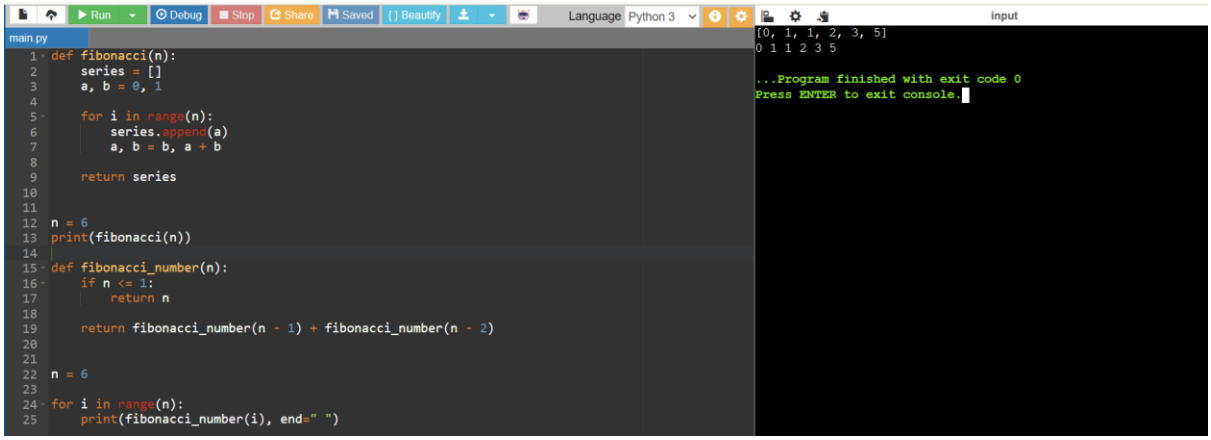
● Problem Statement: Generate first n Fibonacci numbers.

● Sample Input: n = 6

● Sample Output: [0, 1, 1, 2, 3, 5]

● Complexity: Iterative O(n), Recursive naïve O(2n), Recursive with memorization O(n)

● Explanation: Illustrates exponential recursion vs efficient iteration.



Experiment 5: Matrix Multiplication

● Objective: To analyse nested loop complexity.

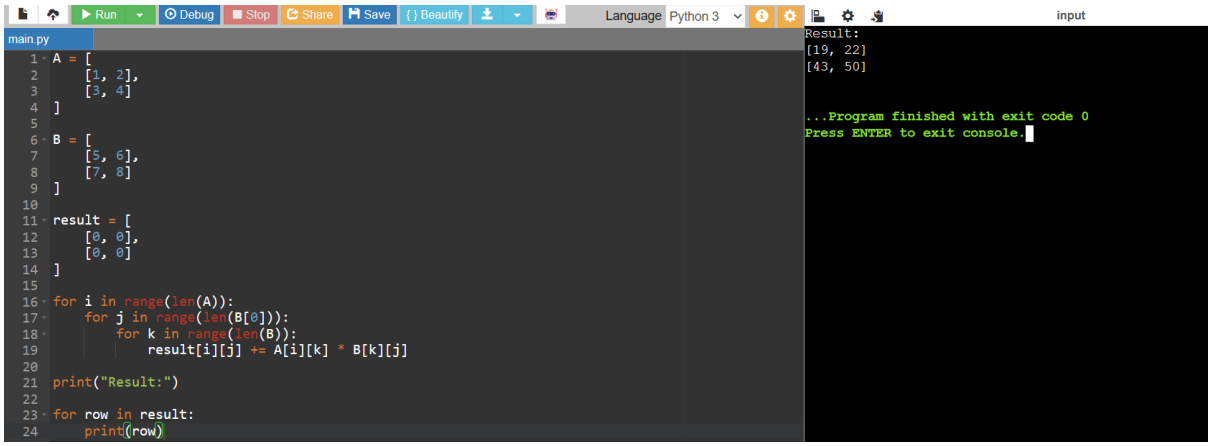
● Problem Statement: Multiply two matrices using loops.

● Sample Input: A=[[1,2],[3,4]], B=[[5,6],[7,8]]

● Sample Output: [[19,22],[43,50]]

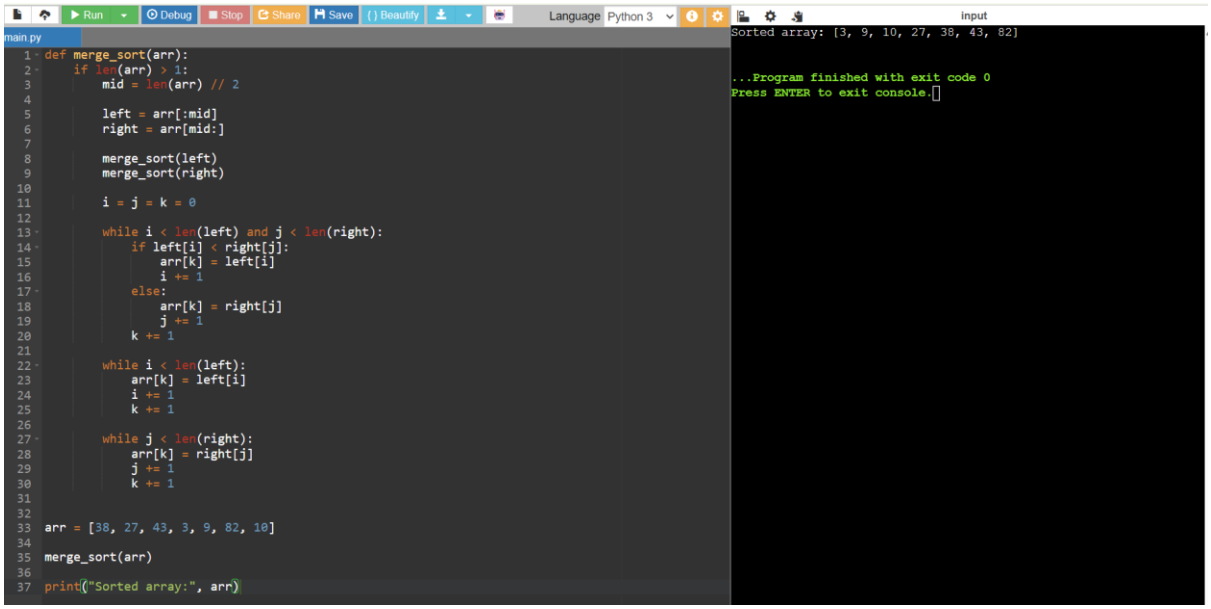
● Complexity: O(n3)

● Explanation: Shows cubic growth due to nested loops.



Experiment 6: Merge Sort

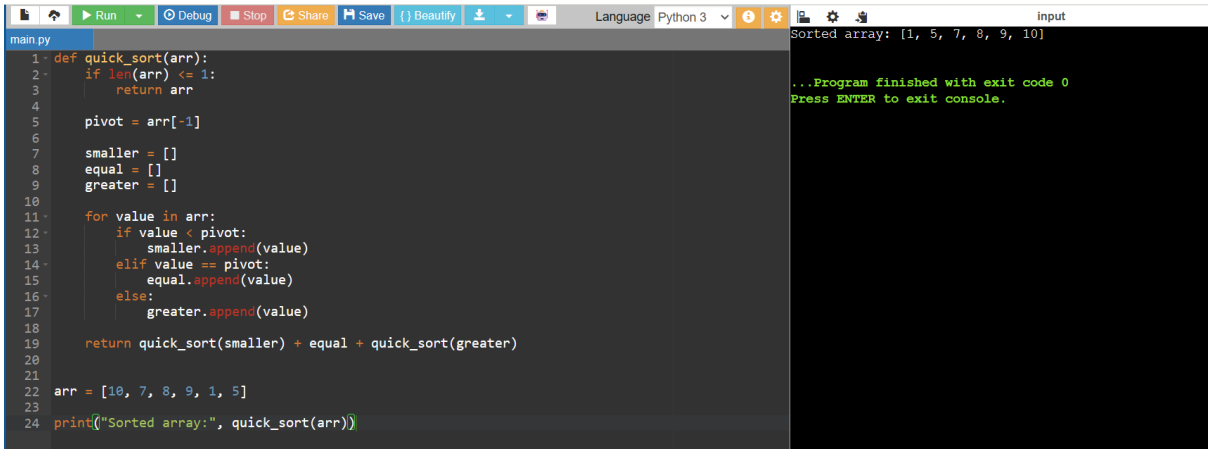
- Objective: To analyse divide-and-conquer sorting.
- Problem Statement: Implement merge sort.
- Sample Input: [38, 27, 43, 3, 9, 82, 10]
- Sample Output: [3, 9, 10, 27, 38, 43, 82]
- Complexity: Always O(nlogf_0n)



- Explanation: Demonstrates guaranteed efficiency of divide-and-conquer.

Experiment 7: Quick Sort

- Objective: To analyse recursive quick sort.
- Problem Statement: Implement quick sort.
- Sample Input: [10, 7, 8, 9, 1, 5]
- Sample Output: [1, 5, 7, 8, 9, 10]
- Complexity: Ω: O(nlogf_0n), O: O(n2), Θ: Θ(nlogf_0n)
- Explanation: Shows how pivot choice affects performance.

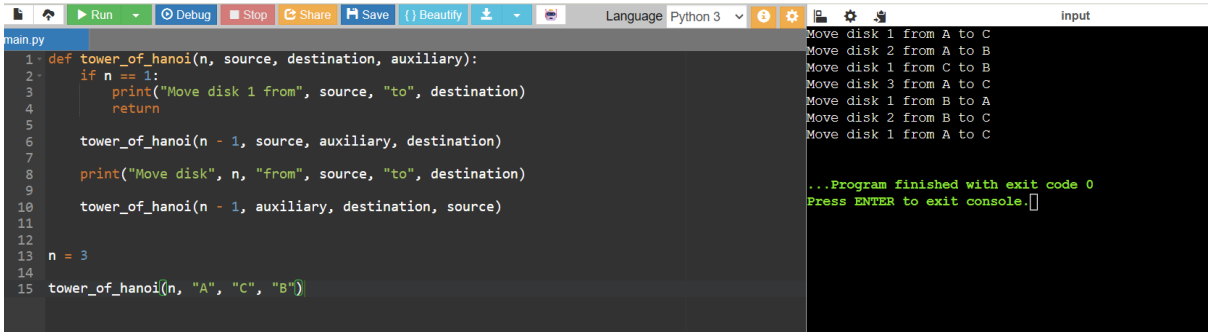


```
def quick_sort(arr):
    if len(arr) <= 1:
        return arr

    pivot = arr[-1]

    smaller = []
    equal = []
    greater = []

    for value in arr:
        if value < pivot:
            smaller.append(value)
        elif value == pivot:
            equal.append(value)
        else:
            greater.append(value)

    return quick_sort(smaller) + equal + quick_sort(greater)


arr = [10, 7, 8, 9, 1, 5]

print("Sorted array:", quick_sort(arr))
```

```
Sorted array: [1, 5, 7, 8, 9, 10]


...Program finished with exit code 0
Press ENTER to exit console.
```

Experiment 8: Tower of Hanoi

- Objective: To analyze exponential recursion.
- Problem Statement: Solve Tower of Hanoi for n disks.
- Sample Input: n = 3
- Sample Output: Sequence of moves.
- Complexity: O(2n)
- Explanation: Illustrates exponential growth in recursive calls.



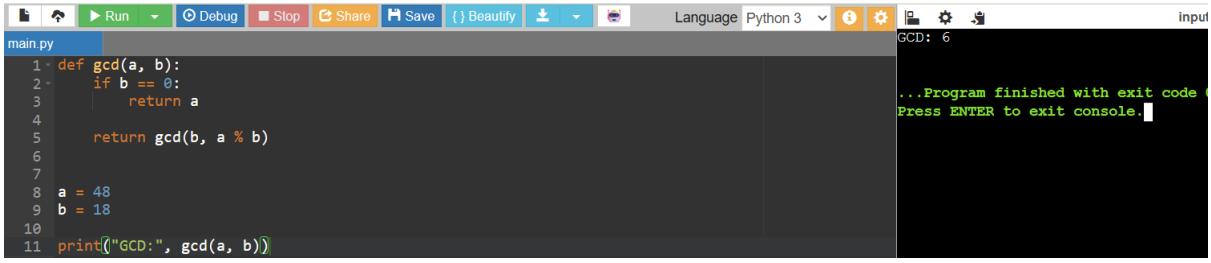
```
def tower_of_hanoi(n, source, destination, auxiliary):
    if n == 1:
        print("Move disk 1 from", source, "to", destination)
        return

    tower_of_hanoi(n - 1, source, auxiliary, destination)

    print("Move disk", n, "from", source, "to", destination)

    tower_of_hanoi(n - 1, auxiliary, destination, source)


n = 3

tower_of_hanoi(n, "A", "C", "B")
```

```
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C


...Program finished with exit code 0
Press ENTER to exit console.
```

Experiment 9: GCD (Euclidean Algorithm)

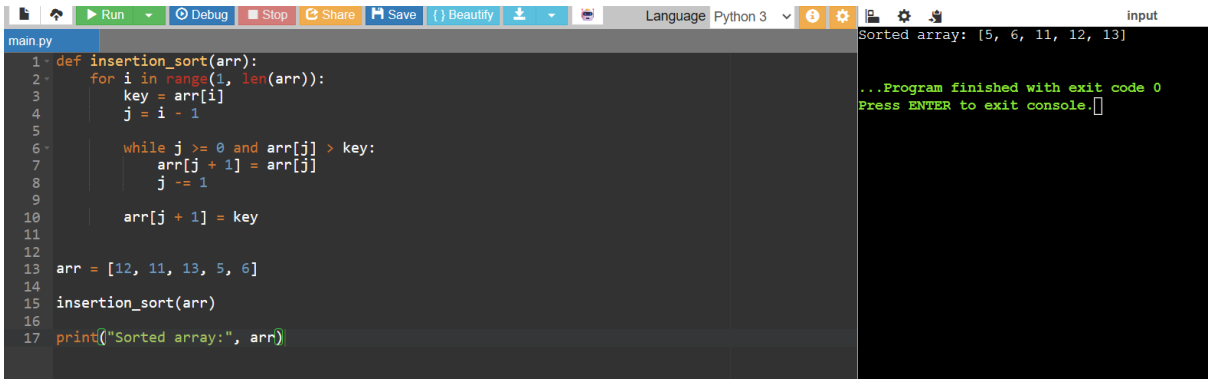
- Objective: To analyze recursive GCD.
- Problem Statement: Implement Euclidean algorithm.
- Sample Input: a=48, b=18

- Sample Output: 6
- Complexity: O(log$_{f0}$n)
- Explanation: Shows efficient recursion with logarithmic complexity.



Experiment 10: Insertion Sort

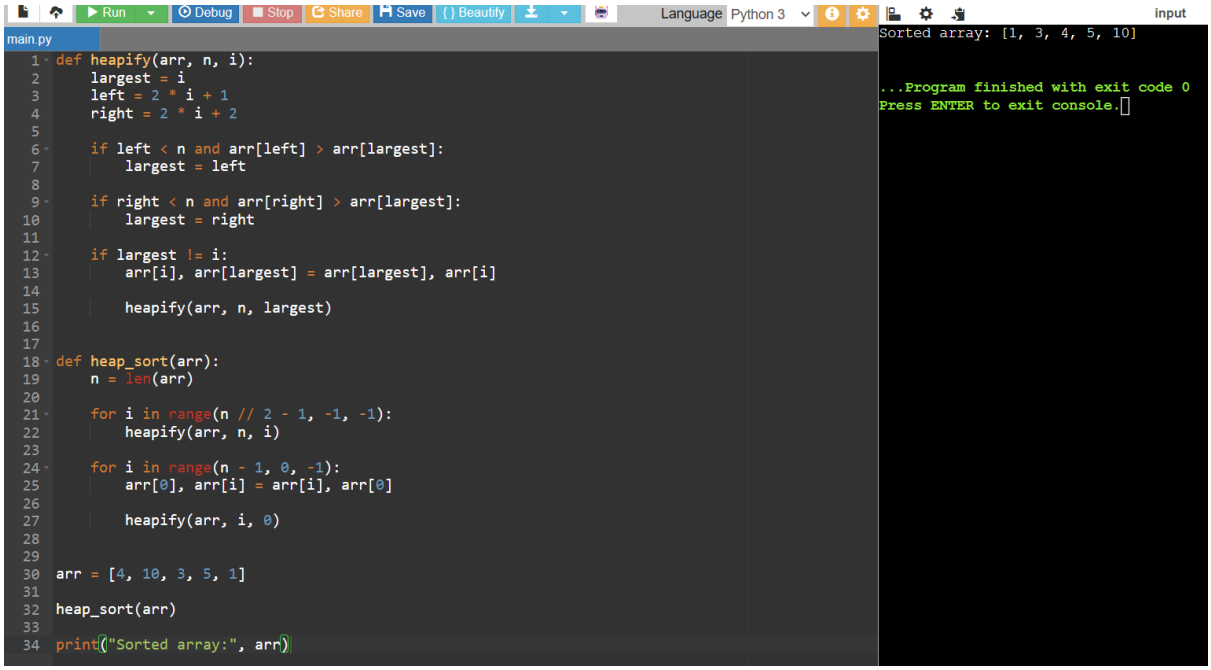
- Objective: To analyze iterative sorting.
- Problem Statement: Implement insertion sort.
- Sample Input: [12, 11, 13, 5, 6]
- Sample Output: [5, 6, 11, 12, 13]
- Complexity: Ω: O(n), O: O(n2), Θ: Θ(n2)
- Explanation: Demonstrates input sensitivity (sorted vs unsorted).



Experiment 11: Heap Sort

- Objective: To analyze heap-based sorting.
- Problem Statement: Implement heap sort.
- Sample Input: [4, 10, 3, 5, 1]
- Sample Output: [1, 3, 4, 5, 10]
- Complexity: Always O(nlog$_{f0}$n)

● Explanation: Consistent performance using heap structure.



Experiment 12: Counting Inversions

● Objective: To analyze modified merge sort.

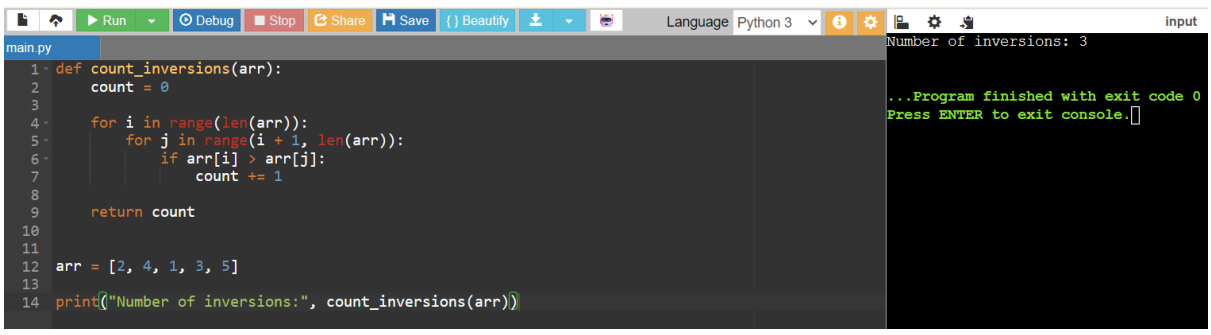
● Problem Statement: Count inversions in an array.

● Sample Input: [2, 4, 1, 3, 5]

● Sample Output: 3

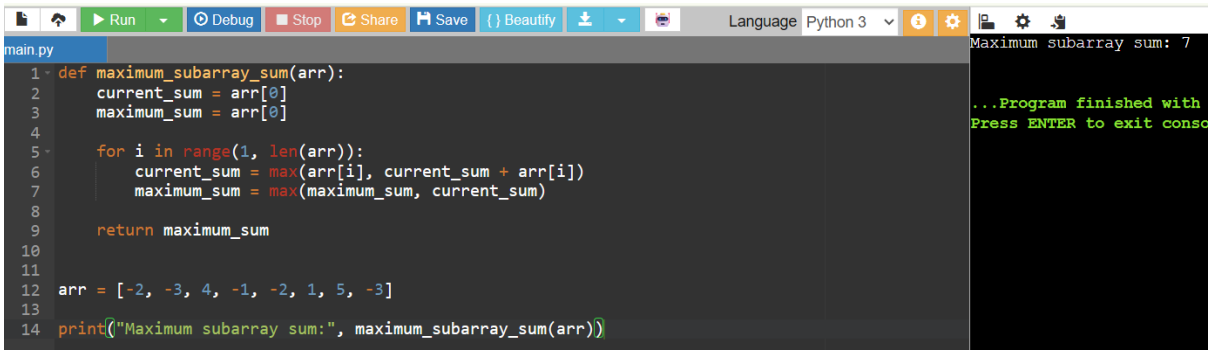
● Complexity: O(nlog[f0]n)

● Explanation: Shows divide-and-conquer applied beyond sorting.



Experiment 13: Maximum Subarray Sum

- Objective: To compare Kadane's vs divide-and-conquer.
- Problem Statement: Find maximum subarray sum.
- Sample Input: [-2, -3, 4, -1, -2, 1, 5, -3]
- Sample Output: 7
- Complexity: Kadane's O(n), Divide & Conquer O(nlogf_0n)
- Explanation: Highlights efficiency differences between approaches.



Experiment 14: Exponentiation

- Objective: To compare iterative vs recursive fast power.
- Problem Statement: Compute xn.
- Sample Input: x=2, n=10
- Sample Output: 1024
- Complexity: Iterative O(n), Recursive fast power O(logf_0n)
- Explanation: Shows how recursion reduces complexity.

```python
def iterative_power(x, n):
    result = 1

    for i in range(n):
        result = result * x

    return result


x = 2
n = 10

print("Result:", iterative_power(x, n))
def fast_power(x, n):
    if n == 0:
        return 1

    half = fast_power(x, n // 2)

    if n % 2 == 0:
        return half * half
    else:
        return x * half * half


x = 2
n = 10

print("Result:", fast_power(x, n))
```

```
Result: 1024
Result: 1024


...Program finished with exit code 0
Press ENTER to exit console.
```

Experiment 15: Closest Pair of Points

- Objective: To analyze brute force vs divide-and-conquer.
- Problem Statement: Find closest pair of points in 2D.
- Sample Input: [(1,2), (4,5), (7,8), (3,1)]
- Sample Output: Closest pair (1,2)-(3,1), Distance = √5
- Complexity: Brute force O(n2), Optimized O(nlogf_0n)
- Explanation: Demonstrates quadratic complexity and motivates optimization.

```python
import math


def closest_pair(points):
    minimum_distance = float("inf")
    pair = None

    for i in range(len(points)):
        for j in range(i + 1, len(points)):
            x1, y1 = points[i]
            x2, y2 = points[j]

            distance = math.sqrt(
                (x2 - x1) ** 2 + (y2 - y1) ** 2
            )

            if distance < minimum_distance:
                minimum_distance = distance
                pair = (points[i], points[j])

    return pair, minimum_distance


points = [(1, 2), (4, 5), (7, 8), (3, 1)]

pair, distance = closest_pair(points)

print("Closest pair:", pair)
print("Distance:", distance)
```

```
Closest pair: ((1, 2), (3, 1))
Distance: 2.23606797749979


...Program finished with exit code 0
Press ENTER to exit console.
```